Contribution Journal 10

Lab: L03_14274_2 Machine Learning Lab:

Course: ITAI1371

Team Members: Katherine Delores, Marisabel Morales, Joshua Balderas

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Overview

For this lab, our group built and evaluated a logistic regression model on the Titanic dataset, focusing on model evaluation through the confusion matrix, classification metrics (accuracy, precision, recall, and F1-score), and cross-validation. Because this was a group project, we divided the required deliverables—the completed notebook PDF, the reflective journal PDF, and this contribution journal PDF—so that each member owned one major piece of the work. Below, I describe in my own words exactly what each team member contributed. As the author of this journal, I also made sure our final files were properly named and clearly formatted as PDFs according to the naming convention: L03_14274-2_ITAI1371.pdf for the notebook, L03Journal_R_14274-2_ITAI1371.pdf for the reflective journal, and L03Journal_C_14274-2_ITAI1371.pdf for this contribution journal.

Katherine Delores – Contribution Journal

My responsibility was the contribution journal—this document. I tracked how the work was divided from the beginning and recorded what each team member did so the final submission accurately reflects everyone's effort. I kept notes on our planning discussions, including why we chose logistic regression as our baseline model and why we relied on k-fold cross-validation instead of trusting a single train/test split. To write accurate descriptions, I coordinated with Marisabel to understand her workflow step by step—her preprocessing decisions, how she generated the confusion matrix, and how she computed the classification metrics—and with Joshua to confirm his reflective journal matched the technical work actually completed. I also checked our deliverables against the lab's self-check list: that all notebook cells were executed with appropriate outputs displayed, that the workflow stages were clearly executed and documented, that the PDF exports captured all content legibly, that Joshua's reflection journal showed genuine reflection on the learning process and concept connections, and that this contribution journal shows the specific contribution made by each student in their own words. Finally, I confirmed the files followed the exact naming convention with our real names. Writing this journal showed me that documentation is not just paperwork—it forces the team to be accountable and makes the project reproducible.

Marisabel Morales – Coding

Marisabel owned the coding portion of the lab and completed it independently from start to finish. She loaded the Titanic dataset and performed all the preprocessing: identifying and handling missing values, encoding categorical variables, and selecting the features used in the model. She then split the data into training and test sets, built the logistic regression model with scikit-learn, and fit it to the training data. After that, she generated the confusion matrix and calculated the classification metrics—accuracy, precision, recall, and F1-score—so we could evaluate the model beyond simple accuracy. She also implemented cross-validation to get a more reliable estimate of performance and compared those results with the single-split evaluation, paying close attention to how data flowed through each workflow step. Throughout the notebook, she added comments explaining each step and produced a visualization of the confusion matrix, which made the results much easier for the rest of the team to interpret. She made sure every cell was executed with outputs visible so the exported PDF clearly demonstrated full workflow understanding. Her clean, well-documented pipeline was the backbone of the entire project.

Joshua Balderas – Reflective Journal

Joshua was responsible for the reflective journal, where he documented the team's learning process and the complexity of the workflow. He wrote about what went smoothly—such as implementing cross-validation and seeing how the metrics gave a fuller picture of model performance than accuracy alone—and what was challenging, including deciding how to handle missing data and understanding why precision and recall can tell different stories about the same model. He reflected on how the confusion matrix helped us see the specific types of errors the model made, and he connected the lab back to the learning types discussed in class, noting when and why different learning approaches are appropriate. Following the lab's tips for success, he related the workflow to real-world examples, paid attention to how data flowed through each step, questioned assumptions such as why cross-validation is more trustworthy than a single split, and explained how the ML tools from Lab 2 fit into this workflow. He also noted possible next steps like hyperparameter tuning or comparing against other models. His reflections tied the technical work back to the course's learning objectives and gave the project a thoughtful conclusion.

Summary

Our group divided the deliverables clearly: I handled the contribution journal and final quality checks against the self-check list, Marisabel built and evaluated the model in the notebook, and Joshua wrote the reflective

journal. Each person's work depended on the others—Marisabel's code and results gave Joshua the material to reflect on, and both gave me the information I needed to document everyone's contributions accurately. This division kept us accountable and ensured the final submission covered the technical, reflective, and documentation sides of the lab, with all files properly named and clearly formatted as PDFs.

This journal was written by Katherine Delores based on her own observations of the team's work.